

PHASE 1 — Purpose Definition

Knowledge Graph of Mongolian Cinema

1. Informal Purpose

When searching for information about Hollywood or Korean films, K-drama online, users can easily access comprehensive information in one place—release year, director, production company, genre, rating, viewer reviews, cast, and their related works.

This is possible because these industries have established * interconnected, structured* knowledge ecosystems supported by mature knowledge graphs.

In contrast, the Mongolian film domain currently lacks such a unified, linked data ecosystem. Information about Mongolian cinema is scattered across many sources, unstructured, inconsistent and disconnected, difficult to integrate or analyze in a single search.

Therefore, this project aims to:

- consolidate dispersed and unstructured Mongolian cinema data,
- build a connected data model using ontology and knowledge-graph methodology,
- provide a comprehensive, searchable, analyzable Knowledge Graph of Mongolian Cinema.

This knowledge graph will enable researchers, film enthusiasts, and media organizations to:

- explore connections among Mongolian films,
- analyze relationships between filmmakers, actors, and studios,
- study genres, award trends, and historical developments,
- access all essential information about a film from a single query,
- perform recommendations and graph-based analytics.

2. Personas

Persona 1: Non-Fan Viewer

Goal: Has no film knowledge. Wants to pick a high-rated film by a director/actor his girlfriend likes for a movie date.

Members: Sainbuyan L., Munknaran B., Altangerel N.

Needs:

- View linked information among director, film, actors
- Runtime and short summary
- High-rated films featuring a preferred artist

Persona 2: Film Enthusiast

Goal: Find other works by a favorite actor or director.

Needs: Similar-genre films, related works by the same creators, ability to discover underrated, lesser-known but high-rated films. Importance placed on rating and release year

Persona 3: Business Owner

Goal: Collaborate with filmmakers popular among target customers (e.g., children, women).

Needs: Networks of artist collaborations, film ratings and recent releases, identify current trending filmmakers with strong portfolios.

3. Scenarios

Scenario A: Non-Fan Viewer

Selecting a good Mongolian film for a date.

KG:

Members: Sainbuyan L., Munknaran B., Altangerel N.

- Actor- list of other high rates films
- Movie- synopsis, runtime
- Director – genre connections to understand film type.

Scenario B: Film Enthusiast

Finding underrated films by a favorite director.

KG:

- Director B – list of all directed films
- Filter by high rating, low review count
- Release year is before 2000

Scenario C: Бизнес эрхлэгч

Сүүлийн үед идэвхтэй, трэнд уран бүтээлчидтай хамтрах

KG:

- Production company - list of produced films
- Filter by 2020–2024 high-rated releases
- Identify frequently collaborating actors and directors

4. Competency Questions (CQs)

The knowledge graph must be able to answer:

1. Can Mongolian films be listed by genre?
2. What films has Director X made, and in which years?
3. Which actors starred in Movie Y?
4. Which Mongolian actors have the highest number of film appearances?
5. Which Mongolian films have won domestic or international awards?
6. What works has Production Company Z released?
7. What common features link films of a specific genre?
8. Can we generate an actor, actor or actor–director collaboration network?
9. Who are the trending filmmakers of the last five years?
10. Can we visualize the career timeline of an actor or director?

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5. Concept Identification

5.1. Core Concepts

Concept	Definition
Movie	Metadata including release year, runtime, genre, rating, summary, awards, review count.
Person	General category for creative individuals (actors, directors, producers, cinematographers, etc.).
Actor	Subtype of Person. Performs roles in films.
Director	Subtype of Person. Oversees artistic and narrative aspects of a film.
Production Company	Entity responsible for financing, production, or distribution.
Genre	Category such as drama, comedy, action, arthouse, etc.
Rating	Numerical quality score (e.g., IMDB score, viewer rating).
Review Count	Number of user reviews or views (used for detecting underrated films).
Release Year	Year the film premiered.
Collaboration	Repeated or significant cooperation between artists (useful for Persona 3).

6. Relationships (Холбоосууд)

Relationship	From → To
Acted in	Actor → Movie
Directed by	Movie → Director
Produced by	Movie → Production Company
Has genre	Movie → Genre
Released in	Movie → Release Year
Has rating	Movie → Rating
Has reviews	Movie → ReviewCount

7. ER Diagram (Conceptual Model)

Entities:

- **Movie:** ID, Name, Release Year, Type, Genre, Rating, Runtime
- **Actor:** ID, Name, Birthdate, Profession
- **Director:** ID, Name
- **Production:** ID, Name

Relationships:

- Movie → Actor (acted_in)
- Movie → Director (directed_by)
- Movie → Production (produced_by)

8. Focus Classification

Concept	Focus Level
Movie	High
Actor	High
Director	High
Genre	Medium
Rating / ReviewCount	High
ProductionCompany	Medium

9. PF-Sheet

Aspect	Definition
Domain	Mongolian cinema (films, cast, crew, studios, genres, ratings)
Problem	Information is scattered, unstructured, and lacks interconnection.
Objective	Create a connected, structured knowledge graph integrating all relevant film data.
Scope	Feature Mongolian films released to date with metadata: title, release year, director, writers, cast, production company, genre, and rating.

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Expected Users	Researchers, film enthusiasts, media organizations, recommendation systems.
Output	Knowledge graph (nodes: movies, actors, directors, production; edges: “acted_in”, “directed_by”, “produced_by”), supporting search and analysis.